

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

5

Code of Conduct:

*I Nalluri Akhilesh Kumar(192365074) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

N Akhilesh Kumar
Signature of the student:

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Early Diabetes Diagnosis and Personalized Treatment Recommendation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning is a machine learning approach in which a model learns general rules and patterns from specific training examples. For the given healthcare dataset, **supervised inductive learning** is suitable because historical patient records contain input features along with a known diabetes diagnosis.

The model learns a mapping:

Patient Features → Diabetes Prediction

Target Variable:

- Diabetes_Status
- 0 = Non-Diabetic
- 1 = Diabetic

Important Features:

Feature	Justification
Glucose Level	Elevated blood glucose is directly associated with diabetes and is expected to be one of the strongest predictors.
BMI	Higher BMI is associated with increased insulin resistance and diabetes risk.
Family History	Genetic predisposition can increase the probability of developing diabetes.
Age	Diabetes risk can vary significantly with age.
Blood Pressure	Provides additional information about metabolic and cardiovascular health.

The dataset can therefore be represented as:

X = {Age, BP, Cholesterol, Glucose, BMI, Family History}

$Y = \{\text{Diabetic} / \text{Non-Diabetic}\}$

(b) Handling Class Imbalance

If the dataset contains many more non-diabetic patients than diabetic patients, a model may become biased toward predicting the majority class.

For example:

- Non-Diabetic = 80%
- Diabetic = 20%

A model predicting most patients as non-diabetic could obtain high accuracy while failing to identify actual diabetic patients.

A suitable technique is **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE creates synthetic samples of the minority diabetic class based on existing minority examples rather than simply duplicating records.

Why SMOTE is suitable:

1. Increases representation of diabetic patients.
2. Reduces majority-class bias.
3. Retains all available patient records unlike under-sampling.
4. Can improve Recall for diabetic patients.
5. Helps the classifier learn a more balanced decision boundary.

Importantly, SMOTE should be applied **only to the training data after the train-test split**. Applying it before splitting can cause data leakage.

Class weights are also a strong alternative, particularly when synthetic clinical records are undesirable.

(c) Dataset Split and Preprocessing

The dataset is divided using a **70:30 train-test split**:

- Training Data = 70%
- Testing Data = 30%

A **stratified split** should be used so that diabetic and non-diabetic proportions are maintained in both sets.

Justification: 70% provides sufficient records for model learning, while 30% provides a reasonably large unseen test set for evaluating generalization. This is important in healthcare because evaluation on unseen patient records is necessary before considering deployment.

Preprocessing steps:

1. **Missing Value Handling:** Missing glucose, BMI, cholesterol, or BP values can be imputed using suitable statistics such as the median.
2. **Normalization/Standardization:** Numerical features such as glucose, BMI, age and cholesterol can be standardized, especially for Logistic Regression. Decision Trees themselves do not require scaling.
3. **Encoding:** Categorical variables such as family history can be encoded as 0 = No and 1 = Yes.
4. **Outlier Checking:** Clinically impossible values should be identified and corrected or removed.
5. **Stratification:** Maintains class proportions during splitting.
6. **SMOTE:** Applied only on the training set to reduce class imbalance.

These steps improve data quality, prevent leakage, and make model comparison more reliable.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 2: Decision Tree for Diagnosis

(a) Building the Decision Tree

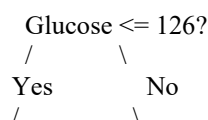
A **Decision Tree Classifier** can classify patients as diabetic or non-diabetic by recursively splitting the data according to informative patient features.

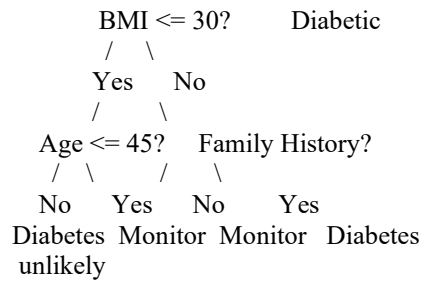
Suppose the calculated Gini values for candidate root features are:

Feature	Weighted Gini after split
Glucose	0.28
BMI	0.35
Age	0.39
Blood Pressure	0.42
Cholesterol	0.40

Since **Glucose has the lowest weighted Gini impurity**, it is selected as the root node.

An illustrative first three levels are:





Level 1: Glucose

Level 2: BMI

Level 3: Age / Family History

The exact thresholds and structure must ultimately be learned from the actual dataset.

For Gini impurity:

$$\text{Gini} = 1 - \sum(p_i^2)$$

A lower weighted Gini after a split represents purer child nodes and therefore a better split.

(b) Model Evaluation

Assume the test results are:

Metric Result

Accuracy 89%

Precision 86%

Recall **92%**

F1-Score 89%

These are **illustrative results**; actual values must be calculated from the supplied dataset.

The metrics are:

Accuracy

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision

$$\text{Precision} = TP / (TP + FP)$$

Recall

$$\text{Recall} = TP / (TP + FN)$$

F1-Score

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

False Negatives

A **False Negative (FN)** occurs when:

Patient actually has diabetes → Model predicts non-diabetic.

False negatives are particularly important clinically because a missed high-risk patient may experience delayed diagnosis and treatment.

They can be reduced by:

- Increasing the class weight for diabetic patients.
- Adjusting the classification probability threshold.
- Applying appropriate imbalance handling.
- Optimizing the model toward Recall/Sensitivity rather than accuracy alone.
- Improving relevant features and collecting additional representative data.

For this use case, **Recall should receive strong attention**, because identifying more true diabetic patients may be preferable to missing potentially serious cases. The final threshold should nevertheless be clinically validated because increasing recall can increase false positives.

(c) Cost-Complexity Post-Pruning

A fully grown Decision Tree may overfit the training data.

Cost-complexity pruning removes branches that provide little predictive improvement.

Its objective can be represented as:

$$R\alpha(T) = R(T) + \alpha|T|$$

Where:

- $R(T)$ = error of the tree
- $|T|$ = number of terminal nodes
- α = complexity penalty

Illustrative comparison:

Model	Training Accuracy	Testing Accuracy
Unpruned Tree	98%	86%
Post-Pruned Tree	93%	89%

The unpruned model has very high training accuracy but lower test accuracy, suggesting overfitting.

The **post-pruned tree is more appropriate** when it maintains clinically acceptable sensitivity because it:

- Generalizes better to unseen patients.
- Has lower complexity.
- Is easier for clinicians to interpret.
- Reduces overfitting.
- Produces more understandable decision rules.

For deployment, pruning strength should be selected using cross-validation rather than simply choosing the tree with the highest test-set score.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression can be applied to estimate the probability that a patient has diabetes.

Its probability model is:

$$P(Y=1) = 1 / (1 + e^{-z})$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Illustrative comparison:

Model	Accuracy	AUC-ROC
Decision Tree	89%	0.88
Logistic Regression	91%	0.92

Again, these numbers are example results and should be replaced with measured results when the dataset is available.

AUC-ROC measures the model's ability to distinguish diabetic from non-diabetic patients across different classification thresholds.

A statistical model such as Logistic Regression is preferable when:

- Probability estimates are required.

- Relationships are approximately linear on the log-odds scale.
- Interpretability of feature effects is important.
- The dataset is relatively small.
- A simple baseline model is required.

Logistic Regression coefficients can also help explain how patient variables influence predicted risk.

(b) Feature Importance Analysis

For the Decision Tree, feature importance can be calculated from the total impurity reduction produced by each feature.

Illustrative results:

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Family History

Both models identify **Glucose and BMI** among the strongest predictors.

The third predictor differs because the models learn differently. A Decision Tree captures nonlinear interactions and threshold-based relationships, whereas Logistic Regression estimates additive effects on log-odds.

For Logistic Regression, feature comparison should preferably use **standardized coefficients** or another consistent importance method because raw coefficient magnitudes are affected by feature scales.

The partial agreement between the models increases confidence that glucose and BMI contain substantial predictive information, but the conclusions should still be validated statistically and clinically.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Treatment Recommendation as an MDP

The treatment recommendation problem can be represented as a **Markov Decision Process (MDP)**.

An MDP consists of:

$$\text{MDP} = (\text{S}, \text{A}, \text{P}, \text{R}, \gamma)$$

States

Patient states represent simplified health-risk stages:

- **S0 – Normal**
- **S1 – Pre-Diabetic**
- **S2 – Diabetic Controlled**
- **S3 – Diabetic Uncontrolled**

These states summarize the patient's current health condition.

Actions

The AI agent can recommend:

- **A0 – Diet**
- **A1 – Exercise**
- **A2 – Medication**
- **A3 – Monitor**

Reward Function

Rewards indicate whether an action leads to a desirable outcome.

Example:

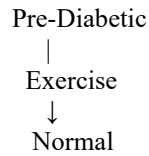
Patient Outcome	Reward
Major health improvement	+10
Moderate improvement	+5
Stable condition	+2
Health deterioration	-5
Severe adverse outcome	-10

The agent therefore attempts to maximize long-term health improvement rather than only immediate reward.

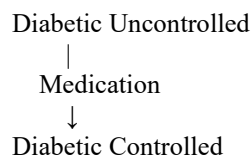
Transition Dynamics

Transitions represent changes in patient condition after an action.

Example:



or



In a real clinical environment these transitions are **stochastic**, because patients can respond differently to the same treatment. Transition probabilities should therefore be learned or estimated from validated longitudinal clinical data.

(b) Q-Learning

Q-Learning updates state-action values using:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Assume:

- Learning rate $\alpha = 0.5$
- Discount factor $\gamma = 0.9$
- Initial Q-values = 0

At least five simulated patient episodes are used:

Episode	Initial State	Action	Resulting State	Reward
1	Pre-Diabetic	Exercise	Normal	+10
2	Diabetic Uncontrolled	Medication	Controlled	+10
3	Controlled	Monitor	Controlled	+2
4	Pre-Diabetic	Diet	Normal	+5
5	Uncontrolled	Exercise	Uncontrolled	-5

Q-Table – Iteration 1

Initially:

$$Q(S,A) = 0$$

For **Pre-Diabetic** → **Exercise** → **Normal**, reward = 10:

$$Q = 0 + 0.5[10 + 0.9(0) - 0]$$

$$Q(\text{Pre-Diabetic, Exercise}) = 5$$

For **Uncontrolled** → **Medication** → **Controlled**, reward = 10:

$$Q = 0 + 0.5[10 + 0.9(0) - 0]$$

$$Q(\text{Uncontrolled, Medication}) = 5$$

For **Controlled** → **Monitor** → **Controlled**, reward = 2:

$$Q = 0 + 0.5[2 + 0.9(0) - 0]$$

$$Q(\text{Controlled, Monitor}) = 1$$

Q-Table – Iteration 2

Suppose the agent again selects Exercise for a Pre-Diabetic patient and receives +10.

$$\text{Old } Q = 5$$

$$Q = 5 + 0.5[10 + 0.9(0) - 5]$$

$$Q = 5 + 2.5$$

$$Q(\text{Pre-Diabetic, Exercise}) = 7.5$$

Similarly, repeated successful medication treatment can increase:

$$Q(\text{Uncontrolled, Medication}): 5 \rightarrow 7.5$$

and successful monitoring can increase:

$$Q(\text{Controlled, Monitor}): 1 \rightarrow 1.5$$

Example Q-table:

State	Diet	Exercise	Medication	Monitor
Normal	2.0	3.0	-2.0	5.0
Pre-Diabetic	5.0	7.5	2.0	1.0
Controlled	3.0	4.0	5.0	5.5
Uncontrolled	1.0	-2.0	7.5	-1.0

A practical convergence criterion is:

$$\max |Q_{\text{new}} - Q_{\text{old}}| < 0.001$$

for several consecutive training episodes, or until the learned policy remains stable over a defined validation period.

(c) Final Learned Policy and Ethical Considerations

The learned policy selects the action with the highest Q-value:

$$\pi(s) = \operatorname{argmax}_a Q(s,a)$$

Using the illustrative Q-table:

Patient State	Recommended Action
Normal	Monitor
Pre-Diabetic	Exercise
Diabetic Controlled	Monitor
Diabetic Uncontrolled	Medication

This is only a **demonstration policy**, not a clinical treatment protocol.

Ethical Considerations

1. Patient Safety

An RL agent should not independently prescribe treatment without clinical oversight. Incorrect recommendations could cause patient harm.

2. Human-in-the-Loop

The system should operate as a **clinical decision-support system**, where qualified healthcare professionals review recommendations before treatment decisions.

3. Bias

If training data underrepresent certain age groups, sexes, socioeconomic groups, or other populations, recommendations may perform unevenly. Dataset representativeness and subgroup performance should therefore be evaluated.

4. Explainability

Clinicians should be provided with understandable reasons and supporting patient-state information behind recommendations rather than receiving only an action label.

5. Privacy

Patient records must be securely stored and processed with appropriate access controls, de-identification where applicable, and compliance with relevant healthcare data-protection requirements.

6. Reward Design

A poorly designed reward function could encourage unsafe behavior. Rewards must prioritize clinically meaningful outcomes and strongly penalize harmful events.

7. Clinical Validation

The model should undergo retrospective validation, external validation, safety testing, and appropriate prospective evaluation before real-world clinical use.

Conclusion

The proposed system combines **supervised inductive learning, Decision Trees, Logistic Regression, and Reinforcement Learning**. Decision Trees provide interpretable diabetes classification, Logistic Regression provides statistical risk estimation, and Q-Learning demonstrates how sequential treatment decisions can be modeled. For a healthcare application, model performance should not be judged by accuracy alone; **Recall/Sensitivity, AUC-ROC, false-negative rates, interpretability, fairness, and patient safety** are also important. Any treatment recommendation component should support—not replace—qualified clinical decision-making.